EXPERIMENT 2

Aim

To convert an ER diagram into a relational schema and implement the schema using MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints.

Theory

An ER diagram is converted into relational tables where entities become tables and attributes become columns. A **primary key** uniquely identifies each record, while a **foreign key** establishes a relationship between two tables. Constraints such as NOT NULL, UNIQUE, CASCADE, and SET NULL maintain data integrity.

Algorithm

1. Identify entities and their attributes from the ER diagram.
2. Convert each entity into a relational table.
3. Select primary keys.
4. Add foreign keys for relationships.
5. Apply required constraints.
6. Insert sample records.
7. Test referential integrity by inserting invalid data.

Program

```
CREATE TABLE Seller (
    seller_id INT PRIMARY KEY,
    seller_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE
);

CREATE TABLE Category (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE Customer (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE
);
```

```
CREATE TABLE Product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    category_id INT NOT NULL,
    seller_id INT NOT NULL,
    FOREIGN KEY (category_id)
        REFERENCES Category(category_id)
        ON DELETE CASCADE,
    FOREIGN KEY (seller_id)
        REFERENCES Seller(seller_id)
        ON DELETE CASCADE
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date DATE NOT NULL,
    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id)
        ON DELETE CASCADE
);

CREATE TABLE OrderItem (
    order_id INT,
    product_id INT,
    quantity INT NOT NULL,
    PRIMARY KEY(order_id, product_id),
    FOREIGN KEY(order_id)
        REFERENCES Orders(order_id)
        ON DELETE CASCADE,
    FOREIGN KEY(product_id)
        REFERENCES Product(product_id)
        ON DELETE CASCADE
);

CREATE TABLE Address (
    address_id INT PRIMARY KEY,
    customer_id INT,
    address VARCHAR(200) NOT NULL,
    FOREIGN KEY(customer_id)
        REFERENCES Customer(customer_id)
        ON DELETE SET NULL
);

INSERT INTO Seller VALUES
(1,'ABC Traders','abc@gmail.com');

INSERT INTO Category VALUES
(1,'Electronics');
```

```
INSERT INTO Customer VALUES
(1,'Rahul','rahul@gmail.com');

INSERT INTO Product VALUES
(101,'Laptop',55000,1,1);

INSERT INTO Orders VALUES
(1001,1,'2026-10-04');

INSERT INTO OrderItem VALUES
(1001,101,2);

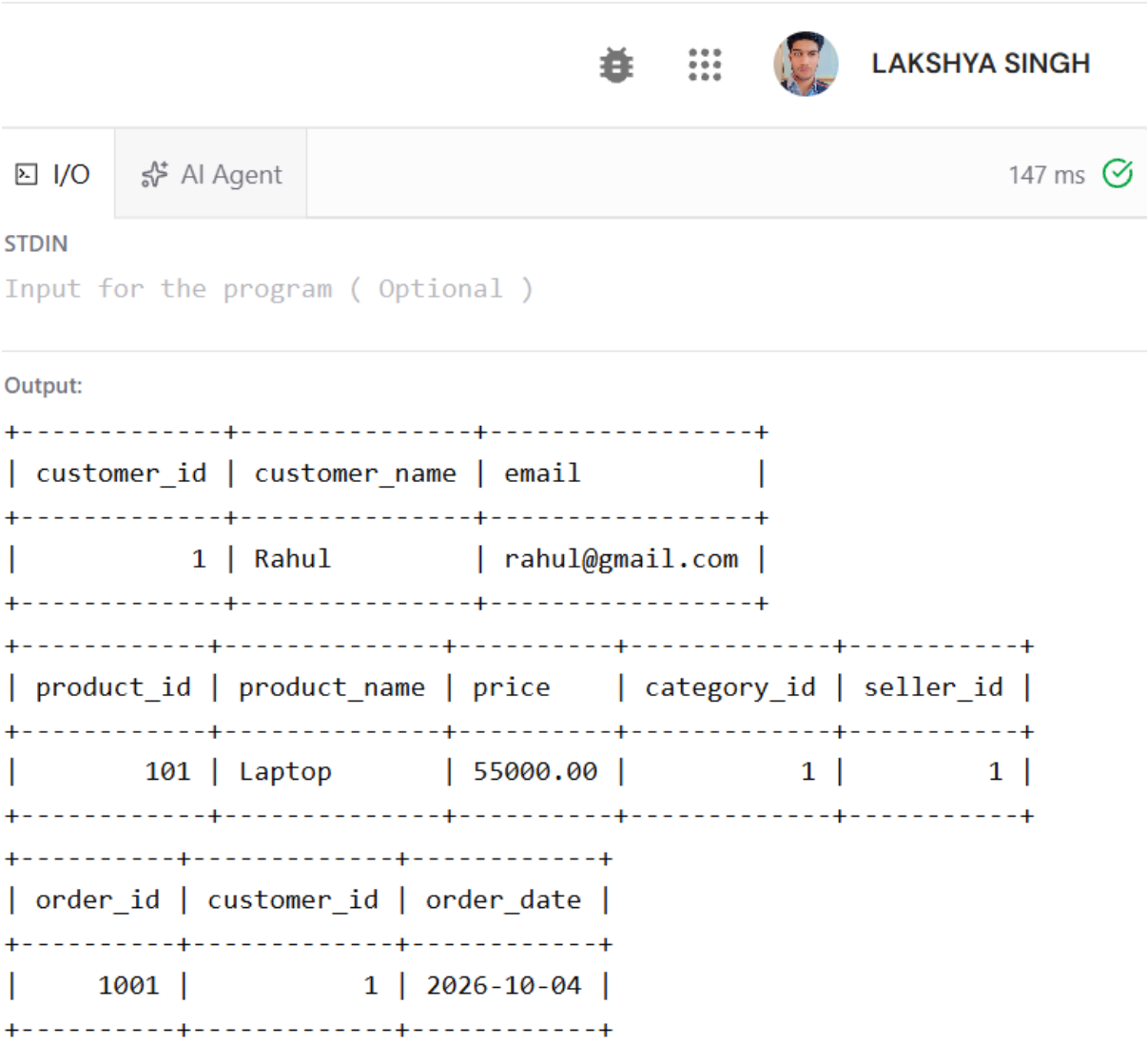
INSERT INTO Address VALUES
(1,1,'Lucknow, Uttar Pradesh');

SELECT * FROM Customer;
SELECT * FROM Product;
SELECT * FROM Orders;
```

Input

Customer, seller, category, product and order details.

Output



Observation

The relational schema was successfully created and the specified integrity constraints were applied.